

TANGO: mapping flood damage from space

Space Track challenge report. Case study: Trishuli flood, Rasuwa, Nepal, 26 August 2026. Repository: <https://github.com/bhoopesh282008/Tango>

Draft, 3 October 2026. The pipeline has run end to end on real Sentinel-1 scenes for the Trishuli area and its output has been compared with the EMSR927 reference (section 5). In short: **what it maps is almost always right, but it finds only a small part of the damage**, between 4% and 35% of the reference area depending on the reach. The flood figures below are therefore lower bounds.

1. What the system does

TANGO takes an area and a flood date and answers three questions for rescue teams: where the flood hit, what was damaged, and who is cut off. It has two parts.

- **A pipeline** (Python, `pipeline/`) that finds a before/after Sentinel-1 pair, calibrates and terrain-corrects it, maps flood water and debris, overlays pre-event OpenStreetMap, and works out which settlements have lost their road connection to a hospital. One command runs it: `python run.py --bbox W,S,E,N --event YYYY-MM-DD`.
- **A dashboard** (React) that shows the result on a map with statistics, a rescue priority ranking, a one-page situation report and a copilot that answers four questions in English and Nepali.

There is no server. The pipeline writes static files and the dashboard reads them.

2. Data and rules

Data	Use	Source
Sentinel-1 GRD (VV, VH)	Before/after radar scenes	Copernicus Data Space
Copernicus DEM GLO-30	Terrain correction, slope, flood path	AWS open data
OpenStreetMap, 27 July 2026	Buildings, roads, bridges, health facilities, places	ohsome API, with Overpass API as fallback
Kuro Siwo	Training the flood model	Hugging Face, CC BY 4.0
Copernicus EMS EMSR927	Checking only	not used as input

No published damage map and no post-event OpenStreetMap edit is used as an input. The OpenStreetMap fetch refuses any snapshot date on or after the event.

One departure from the brief. The ohsome API returned HTTP 403 on its geometry extraction endpoints while its metadata and count endpoints answered normally. The pipeline still asks ohsome first, and when refused takes the same snapshot date from the Overpass API, which serves the same OpenStreetMap history. The output records which service supplied the data. All OpenStreetMap figures in this report came through Overpass.

3. Method

Scene pair. The pipeline searches the Copernicus catalogue 20 days either side of the event and keeps only pairs on the same relative orbit and direction, 12 days apart, with the event between them. Scenes from different tracks view the terrain from different angles and cannot be compared pixel by pixel. Among valid pairs it prefers one whose scenes cover the whole area, then the earliest image after the event. For Trishuli three tracks have a valid pair; it selects track 85 ascending, 16 and 28 August 2026,

whose post-event image is two days after the flood. Track 19 would give one ten days after, and track 121 covers only 43% of the area.

Preprocessing, written from scratch without ESA SNAP. Digital numbers are calibrated to sigma0 with the product's calibration and thermal-noise tables, speckle is reduced with a Lee filter, and the image is terrain-corrected by the Range-Doppler method: for every cell of a 10 m UTM grid the zero-Doppler time is solved on the orbit from the product annotation, the slant range is converted to a ground-range pixel, and the image is sampled there. DEM heights are converted from geoid to ellipsoid first. Cells in layover or shadow are masked. Both scenes are resampled to the same grid, so they are aligned by construction. On the real scenes the result reproduces the product's own geolocation grid to 0.03 pixels, and the geocoded image correlates 0.80 with brightness simulated from the DEM, best at zero shift. Getting there needed one correction found only on real data: the product's slant-to-ground conversion records are not exact at their stated times, and using them as stamped misplaced pixels by up to 57 m.

Flood map. A flood fills valley floors, so change counts as flood only on ground within 30 m above and 600 m of a river (from pre-event OpenStreetMap, or from DEM-derived channels where no river is mapped). There, a drop of more than 3 dB is water or wet sediment and a rise of more than 3 dB is debris. Elsewhere only strong change (4.5 dB) over at least a hectare is kept, as "uncertain". The first version required water to be darker than -18 dB; on the real scenes only 5% of darkened river pixels are that dark, because a river tens of metres wide in a gorge never looks like open water at 10 m. The 30 m and 600 m limits are our assumptions; they were not tuned against any reference map. The alternative method uses the trained model for water (section 4) and the same rule for debris.

Optical evidence. With `--optical` the pipeline adds Sentinel-2. In monsoon most passes are cloudy, so it builds per-pixel composites: the latest clear look before the event and the earliest clear look after it, rejecting cloud, shadow and snow with the scene classification layer. New water is a water index (MNDWI) that turns positive; debris is vegetation (NDVI) that disappears. Optical never removes a radar detection, and follows the same valley-floor rule. Where both agree the confidence is raised; on a valley floor where radar cannot see (layover, shadow) an optical detection is taken at lower confidence; where radar saw nothing and optical finds change, the pixel is marked "uncertain" (off the valley floor only for patches of a hectare or more).

Damage. Buildings, roads and bridges that intersect a water or debris zone are flagged. This is an overlap, not an inspection, so the dashboard labels them "in flood zone" and not "destroyed". Most rural roads have no name in OpenStreetMap, so an unnamed road is listed by its kind and nearest settlement.

Cut-off settlements. Roads become a graph. Flagged road segments are removed. A settlement is cut off when it had a road route to a hospital before the event and has none afterwards. A settlement with no mapped road before the event is reported as "access unknown", so that gaps in the map are not counted as flood damage.

Flood path (bonus). From any point, the path is traced on the DEM by expanding outward and always taking the lowest cell on the frontier. Unlike plain steepest descent, this climbs out of pits and DEM noise. The dashboard runs it in the browser when the user taps the map, and lists the settlements within 500 m of the path in downstream order.

4. AI component: flood segmentation model

A small U-Net (7.8 million weights) labels each pixel as no water, permanent water or flood from four bands: post-event VV and VH, pre-event VV and VH.

Training. Kuro Siwo's labelled Sentinel-1 patches total 170 GB. We trained on a 10 GB sample read as chunks spread evenly over every shard, so that every flood event is represented: 4,240 training patches,

of which 709 patches from three whole events were held out for validation. 40 epochs took about 35 minutes on a laptop GPU (RTX 3050, 6 GB).

Results on the Kuro Siwo test sample, flood class.

Test patches	Count	Model IoU	Model F1	Threshold IoU	Threshold F1
Floods absent from training	312	0.46	0.63	0.27	0.43
Floods also present in training	858	0.69	0.82	0.45	0.62
All	1,170	0.67	0.80	0.44	0.61

How to read this. 11 of the 18 test activations also occur in the training sample, as other areas of the same flood. The first row is therefore the honest "never seen" figure. On those seven floods the IoU ranges from 0.16 to 0.82, and three are at about 0.20 or below. The model beats the threshold rule in every row, mostly by finding more of the flood (recall 0.74 against 0.29 on unseen floods) at the cost of more false alarms (precision 0.55 against 0.81).

Himalayan scenes. Almost none of the Kuro Siwo sample is mountainous: only 36 test patches have more than 300 m of relief, all from floods also present in training. On the real Trishuli scenes, which the model has never seen, it marks 0.53 km² of water on valley floors against 1.62 km² for the threshold rule, and 0.43 km² is common to both. The model fires where the surface became very dark (median -15 dB after the event): a large new dark patch at the Betrawati confluence and parts of the channel below it. It finds nothing in the upper gorge near Rasuwagadhi. Against EMSR927 (section 5) the model does worse than the threshold rule in all three areas: it finds 2% to 14% of the reference area, against 4% to 23%, at similar precision (0.90 to 0.97). The gain it shows on Kuro Siwo does not carry over to this valley.

5. Results for the Trishuli area

Flood and damage map. Scenes: Sentinel-1D, track 85 ascending, 16 and 28 August 2026. The whole run, from scene search to dashboard files, takes about four and a half minutes on a laptop.

	Threshold rule	Model for water
Water or wet sediment	1.62 km ²	0.53 km ²
Debris (same rule in both)	0.89 km ²	0.89 km ²
Buildings in a flood zone	672 of 85,186	154
Road segments in a flood zone	121 of 2,059 (10.8 km inside a zone)	56
Bridges in a flood zone	14 of 192	7
Settlements cut off	23 of 149	14

The valley floor is 2.8% of the area and 7% of the scene is lost to layover and shadow. Within 100 m of rivers the share of pixels that changed by more than 3 dB is about twice the share elsewhere, so the corridor signal is real, but part of it may be ordinary change in a monsoon river between two dates.

Comparison with EMSR927. The reference was used only after the runs, with nothing tuned on it. `python validate.py <run> cache/reference_emsr927` reproduces every figure here. The four reference areas with products are covered by two runs: the main run above, and a second run to the south-west (84.9 to 85.25 E, 27.75 to 28.02 N) made with the same code and settings as a test on another area. For that area the pair rule chose track 19 descending, 24 August and 5 September, because track 85 covers 99% of it and the rule prefers full coverage; its post-event image is ten days after the flood. The second run maps 6.0 km² (954 of 124,059 buildings, 230 road segments, 41 of 176 bridges, 13 of 116 settlements cut off). The reference records the event as a mass movement and outlines the whole debris corridor by

photo-interpretation of 0.3 m to 1 m optical images of 27 to 31 August. Parts the analysts marked "not analysed" are excluded.

Area (run)	Reference	Ours (water + debris)	IoU	Precision	Recall	Model map: IoU / recall
01 Syapru Besi (main)	1.11 km ²	0.04 km ²	0.04	0.88	0.04	0.02 / 0.02
02 Timure (main)	1.29 km ²	0.06 km ²	0.05	0.99	0.05	0.02 / 0.02
03 Bidur, northern 63% (main)	4.09 km ²	1.01 km ²	0.22	0.93	0.23	0.14 / 0.14
03 Bidur, whole (second)	5.89 km ²	1.84 km ²	0.29	0.95	0.30	not run
05 Phosretar (second)	4.79 km ²	1.80 km ²	0.35	0.94	0.35	not run

Damage (threshold map)	Main run: areas 01, 02, 03 north	Second run: areas 03, 05
Reference buildings destroyed, damaged or possibly damaged; those in our zones	2,538; 559 (22%)	2,952; 1,177 (40%)
Buildings we flagged; those next to a reference building	339; 319 (94%)	693; 649 (94%)
Reference road segments destroyed or damaged; those our zones touch	170; 78 (46%)	333; 192 (58%)
Reference road segments with no visible damage; those our zones touch	197; 6 (3%)	522; 36 (7%)
Reference bridges destroyed or damaged; in our OpenStreetMap; flagged	20; 12; 7	18; 10; 10 (Bidur only)

Reading. The map can be trusted where it says "flood" and cannot be trusted where it says nothing. Recall is poor because of what the radar pair shows, not mainly because of hidden slopes: inside the reference corridor only 1% to 7% of pixels are in layover or shadow, but 70% to 80% changed by less than 3 dB between the two dates. Only 15% of the corridor in the two gorge areas, 29% to 35% at Bidur and 41% at Phosretar crosses the 3 dB threshold in either direction, which caps what any rule of this kind can find; recall rises down the valley as the floor widens. Fresh debris over a gravel river bed or bare ground looks much the same to C-band radar at 10 m, while it is obvious in a 0.3 m optical image. Roads do better than buildings because a long segment needs only one detected crossing.

Optical evidence on real scenes. Both areas were run again with --optical. Four Sentinel-2 passes either side of the event gave a clear view before and after for 31% of the main area and 55% of the second. As designed, optical adds almost nothing to the flood classes: it confirms 0.07 km² of radar detections and fills about 0.01 km² of radar blind spots in each run, so the figures above do not change. Its other output is "uncertain" patches where radar saw no change. On valley floors these carry real signal: counting them, recall against the reference would be 0.33, 0.34 and 0.65 in the main run and 0.65 and 0.51 in the second, at precision 0.70 to 0.96. Away from valley floors they are noise, 58 to 82 km² per run, mostly thin cloud and haze the scene classification missed. We have not promoted the valley-floor patches to a flood class, because that rule would have been chosen after looking at the reference. The first real run also exposed a fault, now fixed and tested: optical detections were not limited to valley floors as radar detections are, which added 7 km² of false debris on hillsides.

Other results on real data:

Item	Result
OpenStreetMap, 27 July 2026	85,186 buildings, 2,059 road segments (1,583 km), 192 bridges, 16 health facilities, 149 named settlements
Settlements with a road route to a hospital before the event	84 of 149; the other 65 are "access unknown"
Settlements with a population in OpenStreetMap	1 of 149
DEM	440 m to 6,888 m; spot heights match known values (Dhunchu 2,018 m, Betrawati 617 m)
Flood path from Rasuwagadhi	59.9 km down the Trishuli; median 13 m from the OpenStreetMap river line, 90% within 42 m

With population recorded for one settlement, the dashboard reports mapped buildings instead of people and does not convert one into the other.

6. Dashboard and situation report

The dashboard shows the flood zones, buildings, roads, settlements and infrastructure on a map with a choice of basemaps; headline figures with breakdowns; a rescue priority ranking of cut-off settlements; and export to CSV, GeoJSON and a printable report. The copilot answers "where did the flood hit", "what infrastructure is damaged", "which settlements are cut off" and "which should be rescued first" in English and Nepali, and assembles the four into a situation report.

Every number comes from the maps. One function derives all statistics from the loaded layers; the cards, the ranking, the exports and the copilot read from it. The copilot is not a language model: its answers are templates filled with those figures, so it cannot invent one. Typed questions are matched by keyword to the four topics, and anything else gets a plain "cannot answer".

Missing is not zero. Where the data lacks something the dashboard says so: unknown road access, population not recorded, power lines not assessed. The priority score uses only the factors recorded for every cut-off settlement, so a gap in the data cannot lift one settlement above another.

By default the dashboard opens on a bundled demo dataset, labelled "Demo data", whose geometry and figures are illustrative.

7. Limitations

- **No early warning.** Sentinel-1 returns to the same track every 12 days. The system maps a flood after the next pass; it cannot warn of a glacier collapse or a flood in progress. For Trishuli the earliest usable post-event scene is two days after the event.
- **Optical helps little as built.** Cloud left 31% and 55% of the two areas with a clear view on both sides of the event. Optical patches off the valley floor are mostly false, and those on it are only marked "uncertain" (section 5).
- **Most of the damage is missed.** Against EMSR927 the map finds 4% to 35% of the affected area and 22% to 40% of the affected buildings (section 5). Every count in this report and in the dashboard is a lower bound; a settlement not shown as cut off may still be cut off. Rescue planning must not treat an unmarked area as safe.
- **The comparison is partial.** It covers one event and the four reference areas that have products, on two different scene pairs, so the two runs are not like for like. The 30 m, 600 m and 3 dB limits were not tuned on the reference, and we have not tried to raise recall, because fitting them to EMSR927 would make this comparison meaningless.

- **Pair choice can cost days.** For the second area the rule took an image ten days after the flood over one two days after that covers 99% of the area.
- **Steep terrain.** Radar cannot see slopes in layover or shadow (7% of the scene). In the upper gorge near Rasuwagadhi, where this flood began, almost nothing is detected.
- **Valley floors only.** Flood classes are limited to ground near a mapped river, using limits we chose. A debris flow down an unmapped gully is at best marked "uncertain".
- **The model does worse than the thresholds here** and has no debris class. Debris always comes from fixed thresholds, which can also fire on wet soil, crops, snow and unrelated landslides.
- **Damage is an overlap.** A building in a flood zone is counted whether or not it was damaged; one destroyed outside the mapped zone is not. At 10 m, single houses are not resolved.
- **OpenStreetMap is incomplete.** Footpaths and small settlements may be missing; 65 of 149 settlements have no mapped road. Building counts stand in for population, each building is assigned to the nearest named place, and not every building is a home. Multipolygon buildings are skipped.
- **Road cuts are coarse.** A road segment touching a flood zone is removed whole from the road network (its length in the totals is only the part inside the zone), so some settlements may be reported cut off when a way through exists, and the reverse. Most of the 14 mapped "hospitals" are unnamed and are probably health posts, which makes the cut-off test easier to pass than it should be.
- **The flood path is a drainage line.** It gives direction, not depth, width, timing or reach.
- **The preprocessing is simplified.** Orbit data comes from the product, not precise orbit files; there is no radiometric terrain flattening; only the range part of the noise tables is applied.
- **Nepali text** was written without review by a native speaker.
- **Not an operational tool.** Map tiles come from free public services with no availability guarantee, and nothing here has been checked in the field.

No images of people affected by the disaster are used anywhere in the system or this report.

8. Attribution

Contains modified Copernicus Sentinel data 2026.

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Bountos, N. I., Sdraka, M., Zavras, A., Karavias, A., Karasante, I., Herekakis, T., Thanasou, A., Michail, D., Papoutsis, I. (2024). Kuro Siwo: 33 billion m² under the water. A global multi-temporal satellite dataset for rapid flood mapping. Advances in Neural Information Processing Systems 37.

Reference maps for checking: European Union, Copernicus Emergency Management Service data (EMSR927).